LEARN MATES

(Year 2 / A2 Level)

Magnetic Fields Lecture Notes

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CHAPTER 20
Magnetic Fields

20.1 magnetism

20.1.1 magnets

magnetic effects are commonly seen in *magnets*

a magnet creates a magnetic field that attracts or repels other magnets

➢ polarity of magnets

a magnet has two poles, the *north pole* and the *south pole*

freely suspend a magnet, the north pole points towards earth's geographic north pole

when two magnets are brought near each other, like poles repel, and opposite poles attract

➢ we use **magnetic field lines** to graphically represent how magnetic field permeate space

by convention, fields lines emerge from north pole and go into south pole

density of field lines shows strength of magnetic field

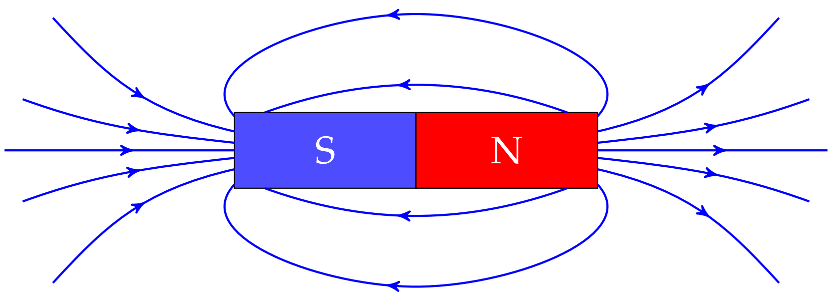
direction of field line tells how the north pole of a small compass will line up at that point

➢ strength of the field is described by a quantity called **magnetic flux density**, denoted by B

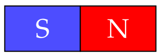
when we draw field lines, we are actually drawing the pattern of flux density B

the notion of flux density will be defined later in details in §20.2.2

Example 20.1 magnetic field around a bar magnet



Question 20.1 For two identical bar magnets placed side by side as shown, what does the magnetic field look like? Try sketching the magnetic field lines.



(a) two attracting bar magnets

(b) two repelling bar magnets

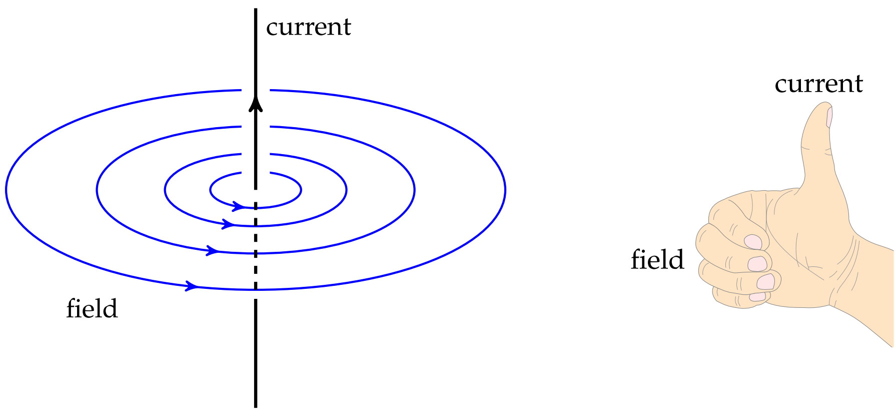
20.1.2 magnetic field due to currents

an electric current also induces a magnetic field around it

this was first discovered by Danish physicist *Hans Christian Orsted* in 1820, when he noticed the turning of a compass needle placed next to a wire carrying current.

we will look into what happens when a current flows through a straight wire or a coil

➢ pattern of the field can be determined using **right-hand (grip) rule**

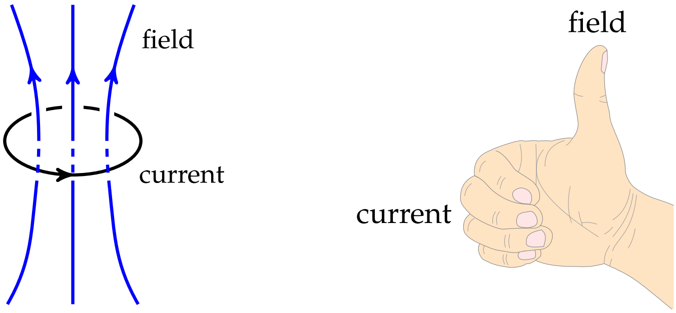


field due to a long straight current-carrying wire

➢ strength of the field is proportional to the current: $B \propto I$

for both straight wires and coils, greater current means stronger field

➢ strength of magnetic field can be increased with *soft iron*



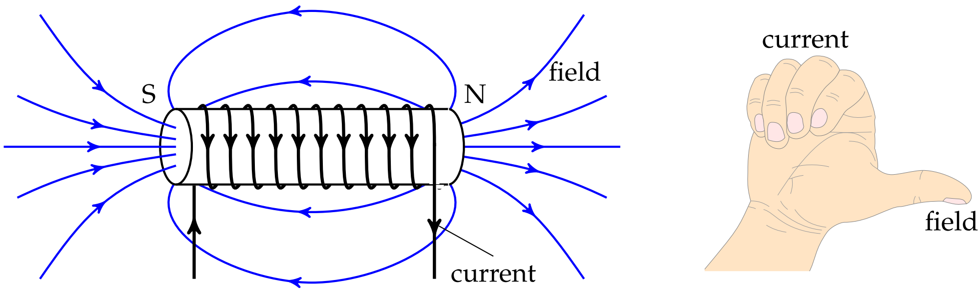
field due to a circulating current

this is because *ferromagnetic materials* (iron, cobalt, nickel) can attract magnetic field lines (with relative permeability μ_r, field becomes $B = \frac{\mu_0 \mu_r I}{2\pi r}$ for wire, $B = \frac{\mu_0 \mu_r N I}{L}$ for coil; iron has large μ_r, greatly intensifying field)

20.1.3 solenoids & electromagnets

strength and polarity of the field due to a coil can be changed easily by tuning currents, so coils are widely used to create magnetic fields where needed

a current-carrying coil is also called a **solenoid**



magnetic field around a solenoid

- a solenoid generates a magnetic field similar to that of a bar magnet

  useful to talk about the north and south poles of a solenoid
- direction of magnetic field in a solenoid is given by the **right-hand (grip) rule**
- magnetic field produced by a solenoid can be controlled

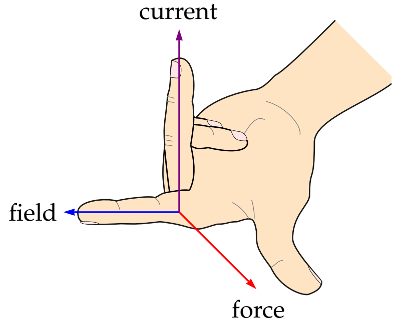
  current $I \nearrow \Rightarrow$ stronger field, also number of turns $N \nearrow \Rightarrow$ stronger field
- inserting an *iron core* inside greatly strengthens the field, this makes an *electromagnet*

Question 20.2 Describe the magnetic field due to an alternating current through a solenoid.

20.2 magnetic force on current-carrying conductor

20.2.1 magnetic force on current-carrying conductor

a current-carrying wire produces its own magnetic field, when it is surrounded by an external magnetic field, the two fields would interact ⇒ a **magnetic force** on the conductor



Fleming's left-hand rule

- direction of magnetic force on current can be worked out with **Fleming's left-hand rule**
- magnitude of magnetic force: $\boxed{F = BIL\sin\theta}$

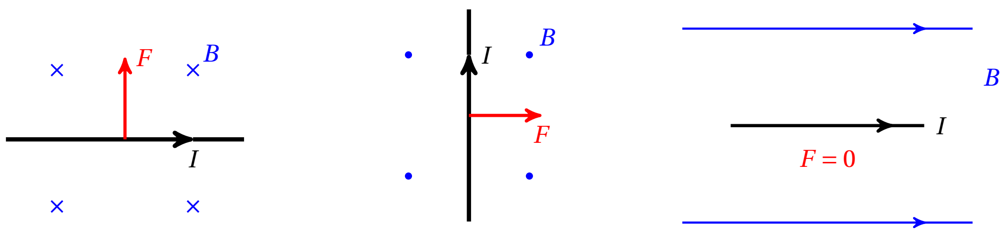
  B is the magnetic flux density to be specified later

  I is the current, θ is the angle between B and I
- when $B \perp I$, magnetic force $F = BIL$

  when $B \parallel I$, there is no magnetic force

  when B forms angle θ with I, only the *perpendicular component* contributes to the force, giving rise to the $\sin\theta$ factor in the formula
- magnetic force is perpendicular to both B and I

Example 20.2 Determine the direction of the magnetic force acting. Check yourself.



(a) (b) (c)

20.2.2 magnetic flux density

rewrite $F = BIL\sin\theta$ as $B = \dfrac{F}{IL\sin\theta}$,we can now give a formal definition for flux density B:

> **magnetic flux density** B at a point is defined as the force acted per unit length on a conductor carrying a unit current at right angle to the magnetic field

➢ flux density describes the strength of a magnetic field (magnetic field strength $H = B/\mu$ is a different quantity)

➢ B is measured in **tesla** (T): $[B] = \mathrm{T}$, $1\ \mathrm{T} = 1\ \mathrm{N \cdot A^{-1} \cdot m^{-1}}$

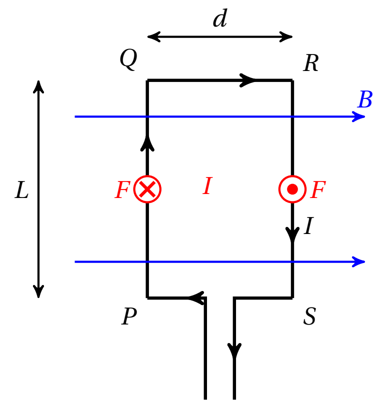
> if a wire of 1 m normal to the magnetic field that carries a current of 1 A experiences a force of 1 N, then the magnetic flux density is 1 T

➢ B is a *vector* quantity, with both magnitude and direction

Example 20.3 A wire of 0.80 m carrying a current of 5.0 A is lying at right angles to a magnetic field, it experiences a force of 0.60 N, what is the flux density?

✍ $B = \dfrac{F}{IL} = \dfrac{0.60}{5.0 \times 0.80} = 0.15\ \mathrm{T}$ (a field of over 0.1 T is actually quite strong!) □

Example 20.4 Torque on a rectangular metal frame in uniform magnetic field



a rectangular frame lies in parallel with the field

$QR, PS \parallel B$, so no force acting on these two sides

$PQ, RS \perp B$, so there is magnetic force $F = BIL$

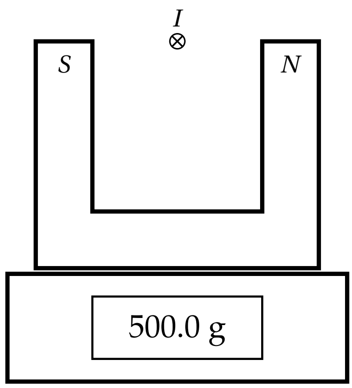
using left-hand rule, we find F_{PQ} acts into the paper, F_{RS} acts out of the paper

this is a pair of equal but opposite forces

they produce a torque about central axis of frame: $\tau = Fd = BILd$, causing the frame to rotate (this effect allows electric motors to be built)

Question 20.3 If the plane of the rectangular frame is at right angle to the magnetic field, describe the magnetic forces acting on each side and hence state what happens to the frame.

Example 20.5 A U-shaped magnet is placed on a balance with a wire suspended above it as shown. Magnetic flux density between the poles is about 0.50 T. The part of the wire that is in the field is of length 8.0 cm. The balance initially shows a reading of 500.0 g. When a current of 10 A flowing into the paper is switched on, what is the new reading on the balance?



✍ use left-hand rule, force on wire acts upwards

from *Newton's third law*, reaction force on magnet acts downwards

so there will be an increase in the balance reading

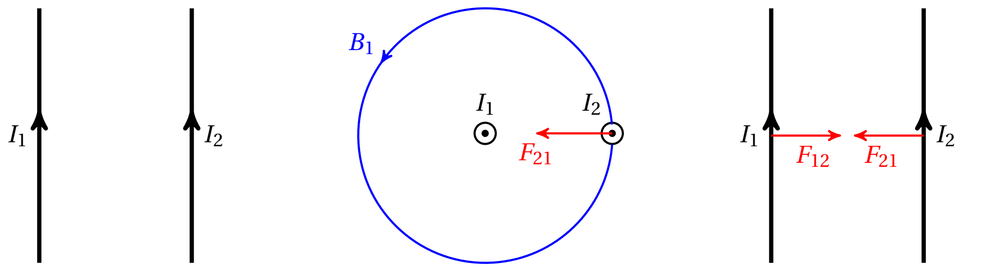
magnitude of the force: $F = BIl = 0.50 \times 10 \times 8.0 \times 10^{-2} = 0.40$ N

change in the mass reading: $\Delta m = \frac{F}{g} = \frac{0.40}{9.81} \approx 0.0407 \text{ kg} \approx 40.7 \text{ g}$

new reading on balance: $m_{\text{new}} = 500.0 + 40.7 = 540.7$ g □

Question 20.4 If the current in Example 20.5 is replaced by a current of 6.0 A flowing out of the plane of the paper. What is the reading on the balance?

force between long straight current-carrying wires



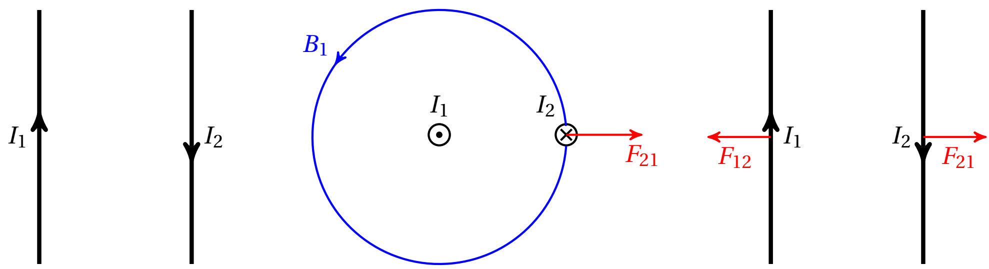
if we have two straight parallel wires carrying currents in the same direction

force acting on I_2 is due to magnetic field B_1 generated by I_1

from top view, I_1 creates a counter-clockwise B_1, so I_2 experiences an upward field

using left-hand rule, force acting on I_2 by I_1, denoted F_{21} (F_{AB} means force on A by B), points to the left

since F_{21} and F_{12} are a pair of action-reaction, so F_{12} points to the right



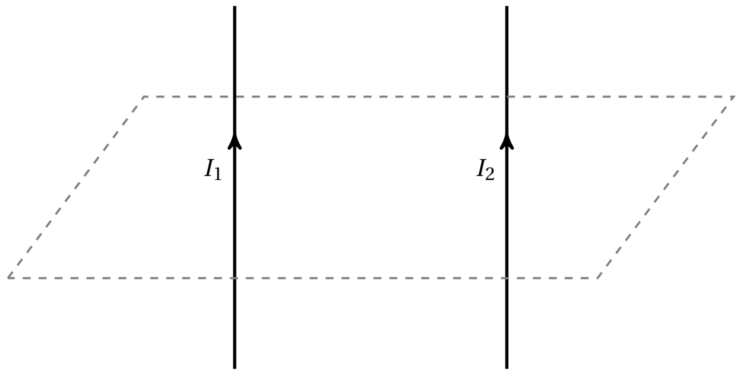
similar discussions apply for I_1 and I_2 flowing in opposite directions (left as exercise)

hence we come to an conclusion: *parallel currents attract, and anti-parallel currents repel* (the ampere is defined by the force between two parallel wires 1 m apart with force 2.0×10^{-7} N/m)

Question 20.5 A light coil of wire of several loops is suspended from a fixed point. When an electric current is switched on in the coil, state and explain the change in the separation between the loops.

Question 20.6 Two long parallel wires carry currents of I_1 =3.0 A and I_2 =5.0 A in the same direction as shown. The flux density at a point at perpendicular distance r from a straight long wire carrying current I is given by $B = \dfrac{\mu_0 I}{2\pi r}$.

(a) Draw at least three lines to show the magnetic field due to I_1.



(b) State and explain the direction of force acting on I_2.

(c) Given that I_1 and I_2 are separated by 4.0 cm, what is flux density due to I_1 at I_2?

(d) What is the magnetic force per unit length acting on I_2?

(e) What is the magnitude and the direction of the force per unit length acting on I_1?

Question 20.7 If the current in one of the two wires in Question 20.6 is replaced by an alternating current, then the two wires should begin to vibrate. However, the wires are not observed to move. Make some reasonable estimates and explain why the vibration is not observed.

20.3 magnetic force on charged particles

electric currents are formed by moving charges, since current-carrying wires in a magnetic field experience force, charged particle moving in magnetic field should also experience force

20.3.1 magnetic force on charged particles

starting from $F = BIl\sin\theta$, let's substitute $I = nAqv \quad \Rightarrow \quad F = B(nAqv)l\sin\theta$

notice n is the number density of charged particles, so nAl together gives the total number

if we look at the force acting on *one* charged particle, then $nAl = 1$, so: $F = Bqv\sin\theta$

➢ magnitude of magnetic force on charged particle: $\boxed{F = Bqv\sin\theta}$

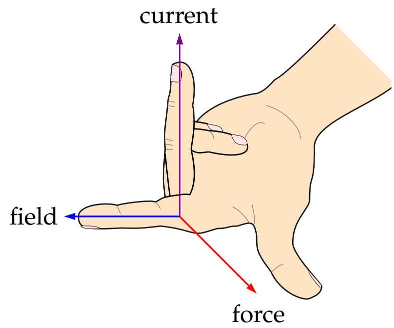
B is magnetic flux density, q is electric charge of the particle, v is its velocity

θ refers to the angle between B and v

➢ direction of magnetic force can be determined using *Fleming's left-hand rule*

for $+q$, current in same direction as v

for $-q$, current in opposite direction as v



Fleming's left-hand rule

➢ force depends on *perpendicular component* of B

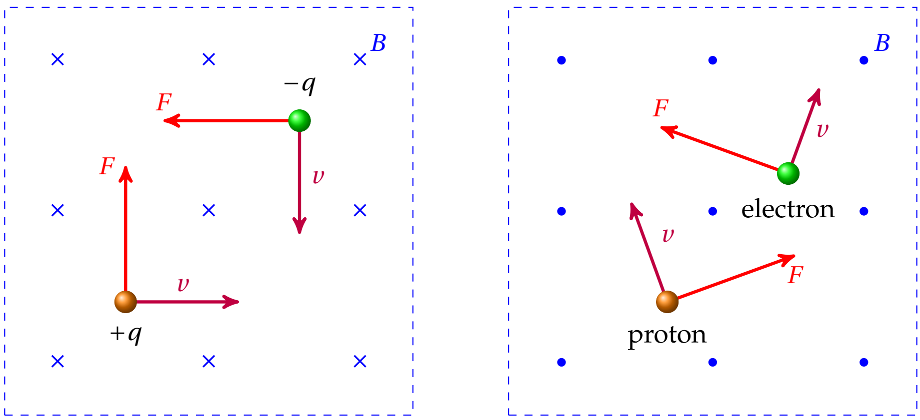
if $v \perp B$, magnetic force $F = Bqv$

if $v \parallel B$, there is no magnetic force

when particle moves at angle θ to B, contribution to the force only comes from the *perpendicular component*, giving rise to the $\sin\theta$ factor

➢ magnetic force always perpendicular to both velocity v and magnetic field B (vector form: $\vec{F} = q\vec{v} \times \vec{B}$)

Example 20.6 Use the left-hand rule to find the direction of the magnetic forces acting on the following moving charges. Check yourself.



(a) (b)

charged particles in electric & magnetic fields

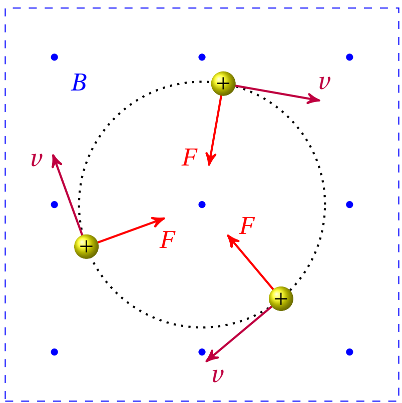
in either electric field or magnetic field, charged particle may experience a force (the combination is the Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$)

in this section, we will compare the difference between electric field and magnetic field

	electric field	magnetic field
strength of the field	electric field strength E	magnetic flux density B
magnitude of force	$F_E = Eq$	$F_B = Bqv\sin\theta$
whether force depends on velocity	no dependence, acts equally on stationary or moving charges	depend on perpendicular component of velocity
direction of force	$F_E \parallel E$ same as/opposite to E for $+q/-q$	$F_B \perp B$ and $F_B \perp v$ use left-hand rule
effect of force on motion of particle	can change both magnitude and direction of velocity	only changes direction of velocity, cannot change magnitude of velocity
work done by force	work can be done, can define notions of potential and P.E.	magnetic force does no work for charged particles

20.3.2 charged particle in uniform magnetic fields

suppose a charged particle is moving at right angle to a uniform magnetic field



circular motion of a charged particle in uniform magnetic field

magnetic force F_B always at right angle to motion, so F_B keeps changing direction of velocity

uniform field means a constant force, so F_B deflects the particle at the same rate

the particle should describe a *circular* path!

for a charged particle moving in a uniform magnetic field, *magnetic force provides centripetal force for circular motion*: $\boxed{Bqv = \frac{mv^2}{r}}$, or $\boxed{Bqv = m\omega^2 r}$

- can solve for radius of the orbiting particles: $r = \frac{mv}{Bq}$
  - $v \nearrow \Rightarrow r \nearrow$, faster particles take larger circles
  - $B \nearrow \Rightarrow r \searrow$, stronger magnetic field, larger centripetal force, smaller circles
  - $m \nearrow \Rightarrow r \nearrow$, larger mass, larger inertia, so larger circles
- radius of curvature relates to charge-to-mass ratio (also called *specific charge*) of the particle
  rearrange the equation we have $\frac{q}{m} = \frac{v}{Br}$, which can be computed using experimental data

Example 20.7 An α-particle travelling at 2.5×10^4 m s^{-1} enters a region of uniform magnetic field. The field has flux density of 5.4 mT and is normal to direction of particle's velocity. What is the radius of α-particle's path?

✍ radius of circular arc: $r = \frac{mv}{Bq} = \frac{4 \times 1.66 \times 10^{-27} \times 2.5 \times 10^4}{5.4 \times 10^{-3} \times 2 \times 1.60 \times 10^{-19}} \approx 0.096 \text{ m} \approx 9.6 \text{ cm}$ □

Question 20.8 For an α-particle and a β-particle entering the same uniform magnetic field at a same speed, compare the radius of their paths.

Question 20.9 For a charged particle undergoing circular motion in a uniform magnetic field, show that its angular velocity is independent of the radius of its path.

Question 20.10 If a charged particle enters a uniform magnetic field at an angle $\theta \neq 90°$, state and explain the path of this particle. (Hint: think about components of its velocity.)

20.3.3 mass spectrometer

a **mass spectrometer** is a device to measure the charge-to-mass ratio of charged particles

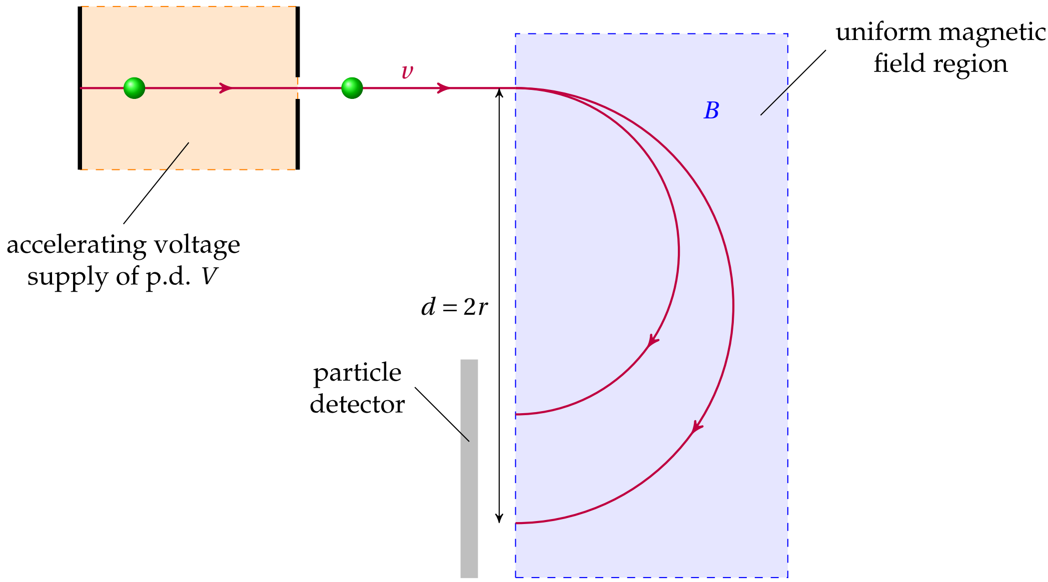
charged particles accelerated through an electric field: $\frac{1}{2}mv^2 = qV$

they then enter a uniform magnetic field: $Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bqr}{m}$

eliminating v: $v^2 = \frac{2qV}{m} = \frac{B^2q^2r^2}{m^2} \Rightarrow \frac{q}{m} = \frac{2V}{B^2r^2}$

we can measure V, B, r in practice, so the charge-to-mass ratio $\frac{q}{m}$ is worked out

different particles have different values of $\frac{q}{m}$, unknown particles can be identified



deflection of two charged particles in a mass spectrometer

Question 20.11 In the figure above, two paths of deflected particles are shown. Give reasons why the radius of the circular path can be different.

Question 20.12 If the particles sent into the mass spectrometer are positively-charged, in which direction should the magnetic field be applied?

Question 20.13 A particle carrying a charge of $+e$ enters a uniform magnetic field of 8.8 mT at right angles with an initial speed of 1.4×10^5 m s^{-1}. It describes a semi-circle with diameter of 66 cm. (a) Find the mass of the particle. (b) Suggest possible composition of this particle.

20.3.4 cyclotron

a **cyclotron** is a type of *particle accelerator*

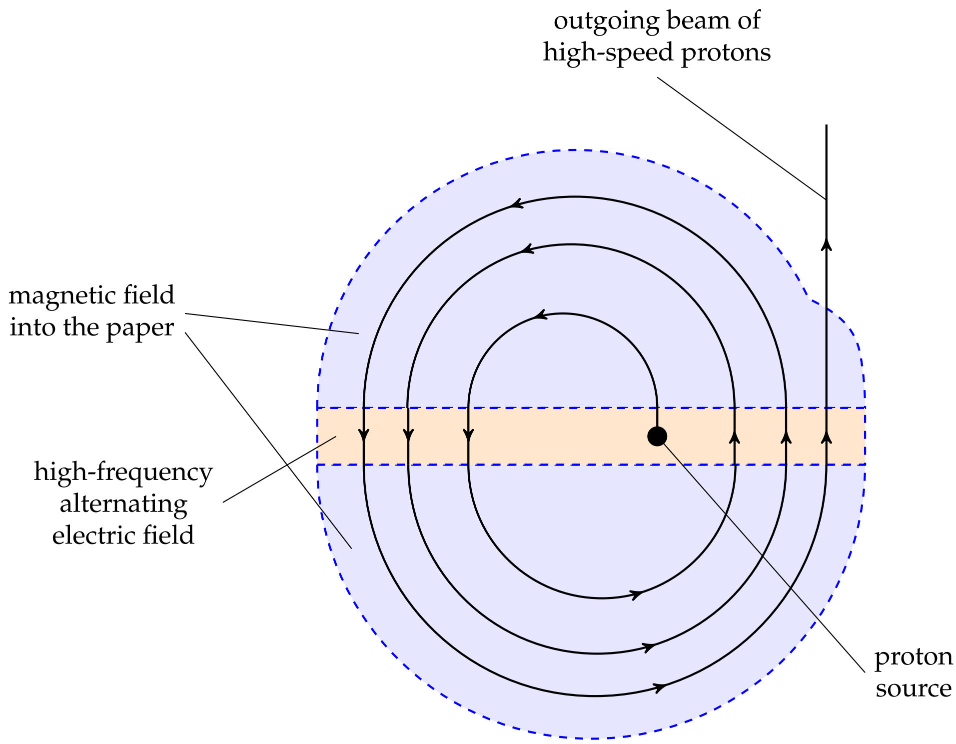
the idea is to make use of a magnetic field to guide moving charges into a spiral path between accelerations by an electric field

as the particle enters and leaves the region of electric field, it gains extra energy of qV

it then follows a semi-circular path under magnetic force and re-enters the electric field

polarity of the electric field is reversed so the particle continues to accelerate across the gap

energy of particle increases by qV, it then moves in a larger semi-circle in magnetic field



protons being accelerated in a cyclotron

repeat this process, the particle leaves the exit port with very high speed

- cyclotron frequency

for charged particles circulating in a uniform magnetic field: $Bqv = m\omega^2 r$

time to complete one full turn is: $T = \dfrac{2\pi r}{v} = \dfrac{2\pi}{v} \times \dfrac{mv}{Bq} \Rightarrow T = \dfrac{2\pi m}{Bq}$

this shows the period is independent of radius of the circular path!

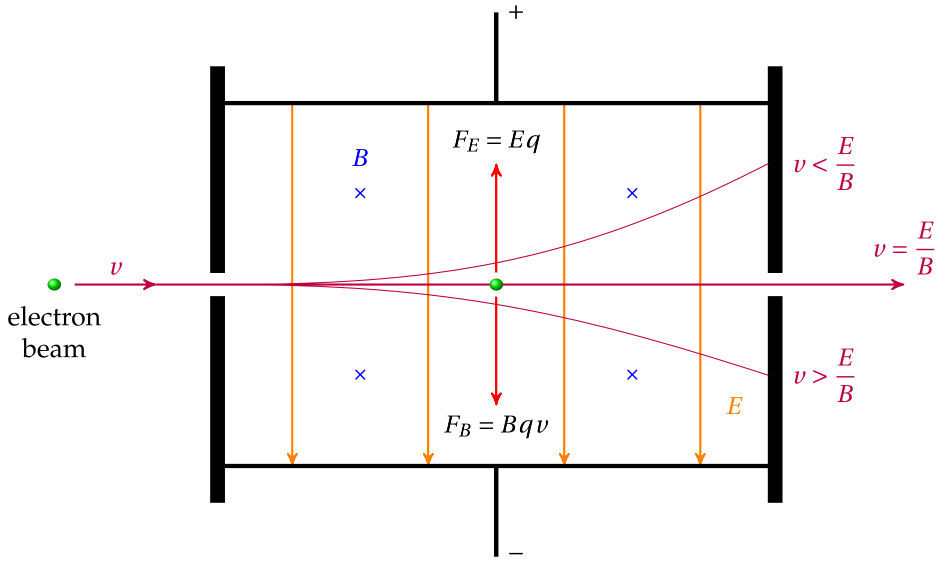
if an alternating voltage is applied at frequency $f = \dfrac{Bq}{2\pi m}$, particles can be accelerated continuously at just the right time when they cross the gap

this frequency is known as the *cyclotron frequency*

Question 20.14 Protons are accelerated in a cyclotron where the magnetic field region has a uniform flux density of 1.5 T and the voltage between the gap is 20 kV. (a) Find the frequency of the voltage supply needed. (b) Find the number of circles the protons must make in order to gain an energy of 10 MeV.

20.3.5 velocity selector

velocity selector is a device to produce a beam of charged particles all with same speed v



velocity selector

let's consider a beam of electrons passing through a region where both uniform electric field and magnetic field are applied, electrons at desired speed v are *undeflected*

no net force acting on these electrons, so equilibrium between electric and magnetic force

$$F_E = F_B \quad \Rightarrow \quad Eq = Bqv \quad \Rightarrow \quad \boxed{v = \frac{E}{B}}$$

for particles entering the same region with higher speed, $F_B > F_E$, they deflect downwards

for particles entering the same region with lower speed, $F_B < F_E$, they deflect upwards

Question 20.15 If electrons at speed v are undeflected when they pass through the velocity selector, what about a beam of α-particles entering the same region at the same speed v?

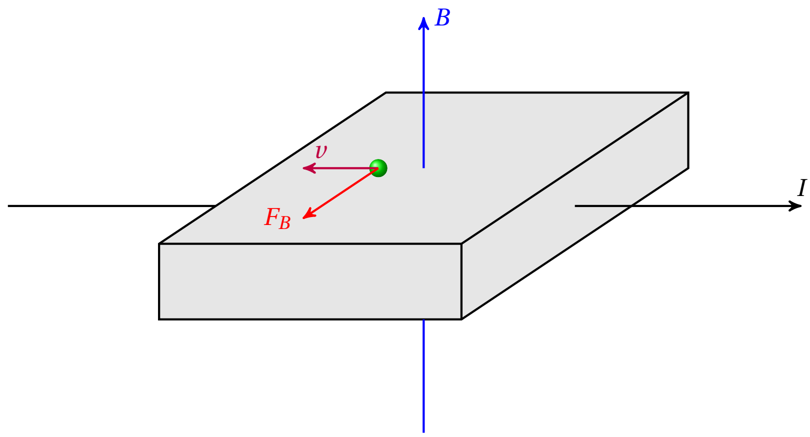
Question 20.16 A uniform magnetic field with flux density 6.0×10^{-2} T is applied out of the plane of the paper. A beam of protons are travelling into this region at a speed of 3.5×10^4m s^{-1}. (a) In which direction do the protons deflect? (b) A uniform electric field is now applied in the same region so that the protons become undeviated. What is the strength and the direction of this electric field?

20.4 Hall effect

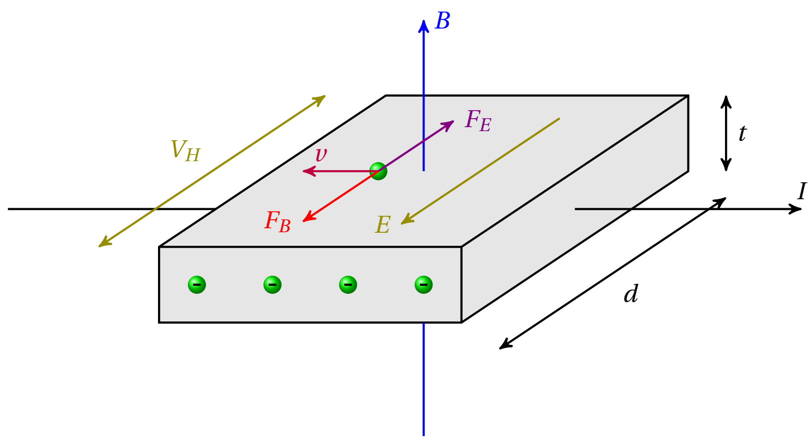
when an electric current passes through a conductor surrounded by an external magnetic field perpendicular to the current, a transverse potential difference is produced

this phenomenon is known as the **Hall effect** (discovered by American physicist *Edwin Hall* in 1879), the voltage difference is called the **Hall voltage**

suppose magnetic field goes upward, current flows to the right (see figure), then charge carriers (assumed to be electrons for now) experience a magnetic force pointing out of paper



this magnetic force causes a build-up of extra charge on front side of conductor, which produces an internal electric field E inside conductor, i.e., a potential difference V_H is formed



as charge carriers accumulate on one side, Hall voltage opposes further migration of charges

steady V_H is established when magnetic force and electric force are balanced: $Eq = Bqv$

- strength of internal electric field is related to Hall voltage: $E = \frac{V_H}{d}$

  where d is width of the conductor

- drift velocity v of particles is related to current I: $I = nAqv$

  where A is cross section of conductor and n is number density of charge carriers

we now have: $\frac{V_H}{d} q = Bq \frac{I}{nAq} \Rightarrow V_H = \frac{BId}{nAq}$

note that cross section $A = dt$, where t is thickness of conductor as shown

we therefore obtain an useful expression for Hall voltage: $V_H = \frac{BId}{n(td)q} \Rightarrow \boxed{V_H = \frac{BI}{ntq}}$

➢ to produce noticeable V_H, we want smaller n and smaller t

- $n \downarrow \Rightarrow$ few free charge carriers $\Rightarrow$ *semi-conductors* are preferable
- $t \downarrow \Rightarrow$ small thickness $\Rightarrow$ *thin slice* of component is preferable

➢ polarity of V_H is determined by nature of charge carries

charge carriers can be *free electrons* (negatively charged) or *holes* (positively charged)

then Hall voltage induced would have opposite polarity

➢ if current I forms angle θ with flux density B, again only perpendicular component matters

expression for Hall voltage becomes: $V_H = \frac{BI}{ntq} \sin\theta$

➢ apply a fixed current in a conductor, V_H is proportional to B

so magnetic flux density can be calculated once we find V_H

one can make use of Hall effect to build a *Hall probe*, a device used to measure flux density B

when using a Hall probe, one should rotate and record the greatest voltage reading to ensure the current applied is at right angle to the external magnetic field

Question 20.17 Why is it difficult to detect Hall voltage in a thin slice of copper?

Question 20.18 A Hall probe is placed near to one end of a strong magnet. State and explain the variation in the Hall voltage as the probe is rotated for one complete revolution.

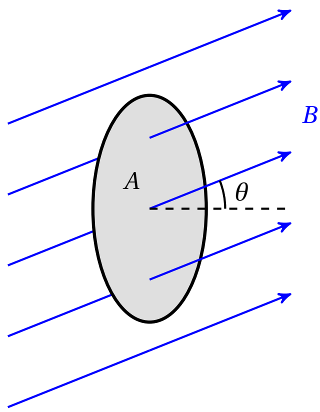
20.5 Electromagnetic Induction

20.5.1 magnetic flux

magnetic flux is defined as the product of a closed area and the magnetic flux density going through it at right angles: $\boxed{\phi = BA\cos\theta}$

- unit for magnetic flux: $[\phi] = \text{Wb}$, $1\ \text{WB} = 1\ \text{T}\cdot\text{m}^2$
- if magnetic field is perpendicular to the area

  magnetic flux simply becomes: $\boxed{\phi = BA}$



- for a coil with N turns, total magnetic flux is

  $$\boxed{\Phi = N\phi = NBA}$$

  this is called **magnetic flux linkage**
- magnetic flux ϕ can be graphically thought as the number of field lines through area A

  cutting of field lines means a change in flux
- change in flux can be caused by many processes, for example
  - moving a magnet towards/away from a coil
  - varying current in a solenoid/electromagnet
  - inserting an iron core into a solenoid/electromagnet
  - pushing a straight conductor through a magnetic field
  - rotating a coil in a magnetic field
  - …

Question 20.19 Explain why the processes mentioned above would give rise to a change in magnetic flux.

20.5.2 laws of electromagnetic induction

electromagnetic induction is the phenomena that magnetism produces electricity

the laws of electromagnetic induction were discovered by *Michael Faraday* and *Heinrich Lenz* in the 1830s, and later described mathematically by *James Clerk Maxwell*

we will study in what conditions voltages and currents could be induced, and how to find their magnitudes and polarities

20.5.3 Faraday's law

Faraday's law states that induced e.m.f./voltage is proportional to rate of change in magnetic flux (linkage): $\boxed{\mathscr{E} \propto \dfrac{\Delta\Phi}{\Delta t}}$

- Faraday's law gives the *magnitude* of the induced e.m.f.

  the key here is the *change* in flux: as long as flux changes, e.m.f. will be induced
- if $\mathscr{E}$, Φ, t are all given in SI units, this proportional relation becomes an identity: $\boxed{\mathscr{E} = \dfrac{\Delta\Phi}{\Delta t}}$

Example 20.8 A coil of 80 turns is wound tightly around a solenoid with a cross-sectional area of 35 cm^2. The flux density at the centre of the solenoid is 75 mT. (a) What is the flux linkage in the coil? (b) The current in the solenoid is *reversed* in direction in a time of 0.40 s, what is the average e.m.f. induced?

✍ flux linkage: $\Phi = NBA = 80 \times 75 \times 10^{-2} \times 35 \times 10^{-4} = 0.21$ Wb

average e.m.f. induced: $\mathscr{E} = \dfrac{\Delta\Phi}{\Delta t} = \dfrac{(+\Phi) - (-\Phi)}{\Delta t} = \dfrac{2 \times 0.21}{0.40} = 1.05$ V □

20.5.4 Lenz's law

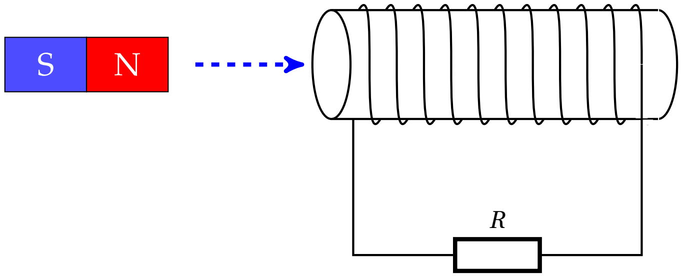
Lenz's law states that induced e.m.f. or current is always in the direction to produced effects that *oppose* the change in flux that produced it

- Lenz's law gives the *polarity* of the induced current

  the key here is the nature dislikes any change in flux: effects of induced current always *oppose* the cause of its production
- Lenz's law is a consequence of the conservation of energy

  since induced current dissipates electrical energy as heat, it must cause loss of original forms of energy possessed by the system if the change in flux is caused by motion, then magnetic force on/by the induced current must *resist* this motion

Example 20.9 A coil of 200 turns is connected to a resistor $R = 4.0\ \Omega$. The coil has a diameter 10 cm. Initially a bar magnet is at great distance from the coil. The magnet is then inserted into the coil and field inside coil becomes 0.40 T. The process occurs within a duration of 2.0 s.



(a) What is average induced e.m.f. in the coil? (b) What is average induced current through resistor R? (c) In which direction does this current flow?

✍ change in flux linkage: $\Delta\Phi = NBA - 0 = 200 \times 0.40 \times \pi \times 0.050^2 \approx 0.628$ Wb

average e.m.f. induced: $\mathscr{E} = \dfrac{\Delta\Phi}{\Delta t} = \dfrac{0.628}{2.0} \approx 0.314$ V

average current induced: $I = \dfrac{\mathscr{E}}{R} = \dfrac{0.314}{4.0} \approx 0.0785$ A

as magnet approaches, coil experiences an increasing flux to the right

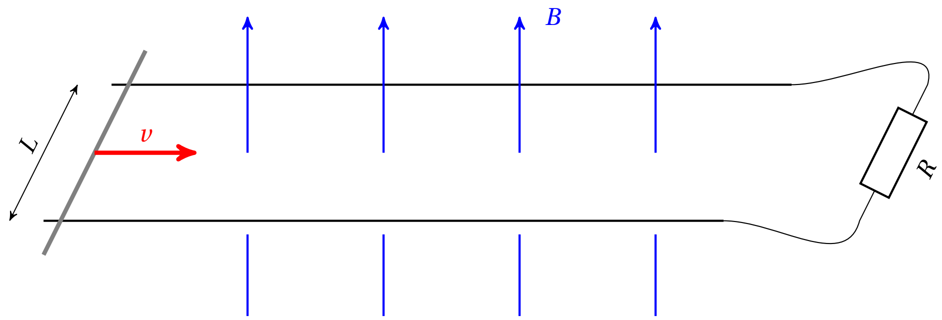
according to Lenz's law, field due to induced current acts to the left to oppose the change

alternatively, left side of the coil should behave as a north pole to oppose motion of magnet

use right-hand rule, we find induced current flows through R to the right

Question 20.20 If the magnet is now pulled away from the coil, state ad explain the direction of the induced current.

Example 20.10 Two parallel metal tracks are separated by a distance of $L = 45$ cm and placed in a uniform magnetic field of 0.80 T. The tracks are connected to a resistor $R = 6.0\ \Omega$. A long metal rod is being pushed under an external force at $v = 3.0$ m s^{-1} along the tracks as shown.



(a) What is the induced current through resistor? (b) In which direction does this current flow? (c) For the rod to travel at constant speed, what is magnitude of external force required?

✍ change of flux in time interval Δt is: $\Delta\phi = \Delta(BA) = B\Delta A = BLv\Delta t$

e.m.f. induced: $\mathscr{E} = \dfrac{\Delta\phi}{\Delta t} \Rightarrow \boxed{\mathscr{E} = BLv}$

induced current: $I = \dfrac{\mathscr{E}}{R} = \dfrac{BLv}{R} = \dfrac{0.80 \times 0.45 \times 3.0}{6.0} = 0.18A$

according to Lenz's law, magnetic force on induced current opposes the rod's motion of cutting field lines, so magnetic force acts to the left

using Fleming's left-hand rule, we find induce current flows in anti-clockwise direction

if rod travels at constant speed, then equilibrium between external push and magnetic force

external force required: $F_{\text{ext}} = F_B = BIL = 0.80 \times 0.18 \times 0.45 \approx 0.065$ N □

Question 20.21 In Example 20.10, can you find alternative arguments to determine the direction of the current induced as the rod cuts through the magnetic field?

Hall voltage & induced voltage in a coil

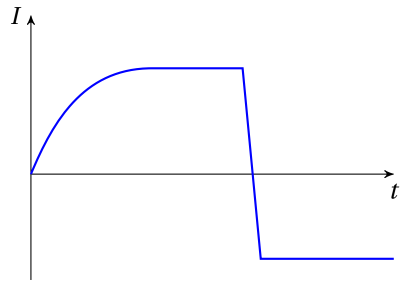
in this section, we compare readings on voltmeter connected to a Hall probe and a coil

Hall probe picks up voltage proportional to flux density: $V_H \propto B$

coil picks up induced e.m.f. proportional to *rate of change* in flux: $\mathscr{E} \propto \dfrac{\mathrm{d}\Phi}{\mathrm{d}t}$

Question 20.22 If a Hall probe is placed near a permanent magnet, what voltage do you measure? If a small coil is placed in the same field, state and explain whether you can measure a non-zero voltage in the coil. If not, state three different ways in which you can produce a voltage.

Question 20.23 A Hall probe is placed near one end of a solenoid that carries a varying current as shown in the graph. (a) Sketch the variation of Hall voltage with time. (b) If Hall probe is replaced by a small coil parallel to the solenoid's end, sketch the variation of the voltage induced in the coil.



20.5.5 applications of electromagnetic induction

20.5.6 magnetic braking

induction brakes make use of induced current to dissipate kinetic energy of a moving object

in this section, we will look at two simple demonstrations, but the principle can be used in braking system of high-speed trains, roller-coasters, etc.

the damped pendulum

a metal disc can swings freely between the poles of an electromagnet

when the electromagnet is switched on, the disc comes to rest very quickly

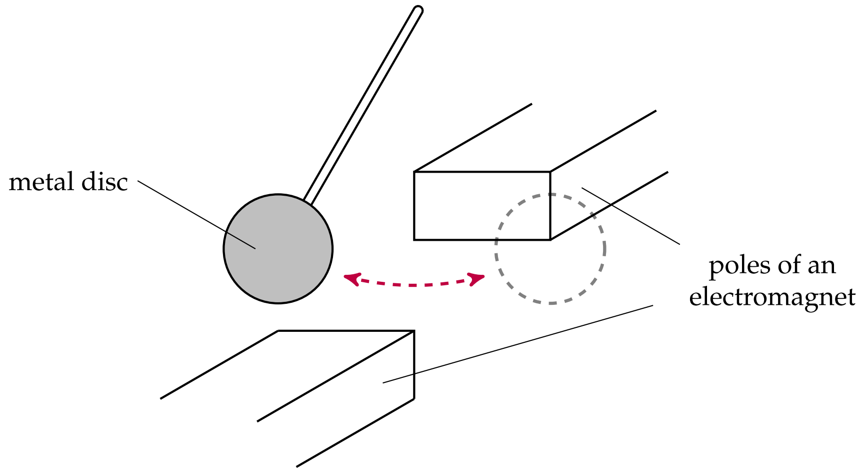
as disc moves in and out of field, change in flux gives rise to e.m.f induced

note that different parts of the disc experience different rates of change in flux, different e.m.f. is induced in different parts for the disc

this causes circulating currents flowing in the disc, called **eddy currents**

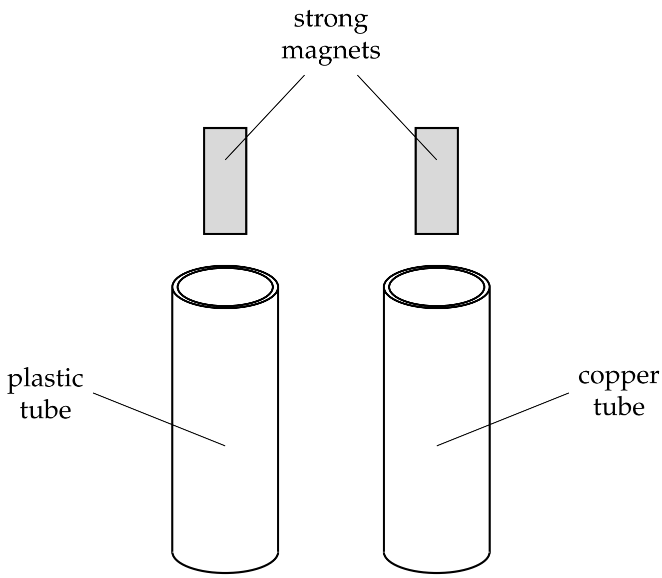
- vibrational energy is lost as heat due to the eddy currents induced
- magnetic force on induced current causes damping

so amplitude of oscillation decreases quickly



falling magnet

suppose we drop two strong magnets down a plastic tube and a copper tube



as magnet falls, tube experiences change in flux, so e.m.f is induced

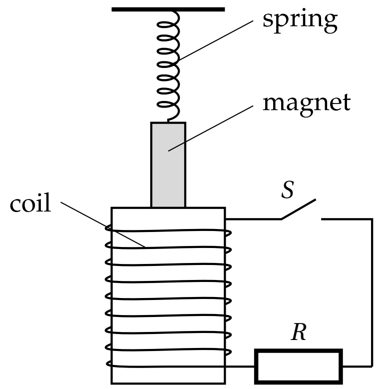
since plastic is an insulator, so no current flows in plastic tube, magnet undergoes free fall

but in the conducting copper tube, eddy current is induced in the tube

- energy is lost as heat due to induced current, not all G.P.E. converted into K.E.
- induced current exerts magnetic force on magnet to oppose its motion, acceleration $a < g$

so magnet falls more slowly through the copper tube

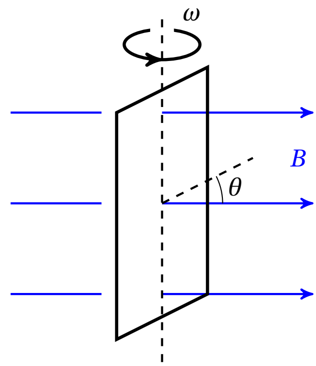
Question 20.24 A magnet is suspended from the free end of a spring. When displaced vertically and released, the magnet can oscillate in and out of a coil (see diagram). The switch S is initially open, there is negligible change in the amplitude. However, when the switch is closed, the amplitude is seen to decrease quickly. Explain the reasons.



the generator imagine a coil rotates with constant angular speed ω in a uniform magnetic field B

let's assume that the coil initially lies in parallel to the field, i.e., $\theta = 0$ at $t = 0$

at time t, coil forms an angle $\theta = \omega t$ with the magnetic field (see diagram)



magnetic flux linkage through coil is:

$$\Phi = NBA\sin\theta = NBA\sin\omega t$$

magnitude of induced e.m.f. is:

$$\mathscr{E} = \frac{\mathrm{d}\Phi}{\mathrm{d}t} = \omega NBA\cos\omega t$$

this is a sinusoidal voltage with maximum value: $\mathscr{E}_{\text{max}} = \omega NBA$

for a coil rotating in a magnetic field like this, an *alternating current* is generated

this is basically how a *generator* works

in practice, it is a strong electromagnet rotating inside a large coil to generate electricity

Question 20.25 State and explain the effect on the voltage generated if the coil rotates faster.

Question 20.26 For the coil rotating at a uniform angular speed in a uniform magnetic field, is the magnetic flux in phase with the e.m.f. generated? If not, what is the phase difference?

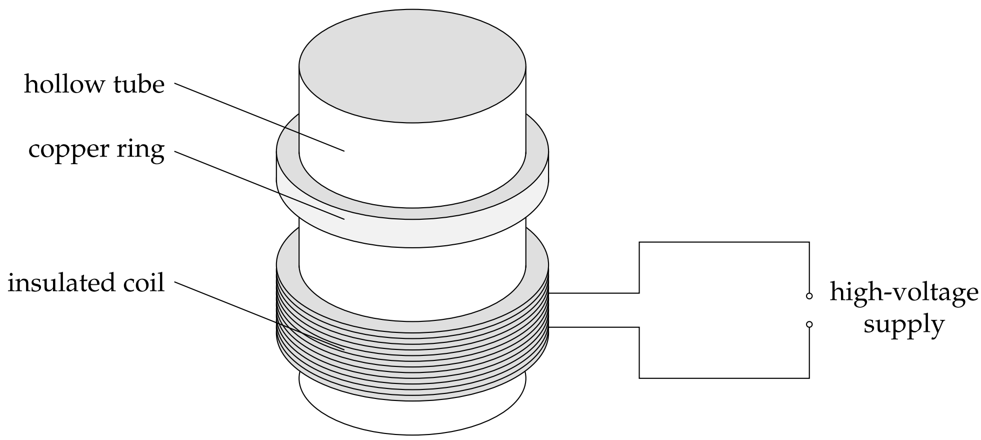
Question 20.27 Does the generator output the maximum voltage when the rotating coil is in parallel to the magnetic field or when the coil is at right angle to the field?

20.5.7 electromagnetic gun

a coil of wire of many turns is wound on a hollow tube

a light copper ring that can move freely along the tube is placed on the coil

let's find out what happens to the ring when a high-voltage supply is switched on



when supply is switched on, flux density in coil greatly increases

sudden change in magnetic flux could induce a huge e.m.f. in copper ring

induced current in ring would experience a repulsive magnetic force

this repulsive force could be way larger that ring's weight, ring would jump out from tube

if the ring is replaced by a small metal sphere placed inside the tube, the same strong magnetic force arising from a sudden change in flux could fire the sphere at very high speed

Question 20.28 The coil is now connected to a stable d.c. voltage supply. If we quickly insert an iron core into the tube, what might happen to the light copper ring?

other applications

as a final remark, electromagnetic induction is widely used in many other areas as well

for those who are interested, you may research on the principles behind wireless charging, induction cooking, contactless payment technology, smart pencils for tablets or computers, etc.